

ANALYTICS SERVICES

Definition

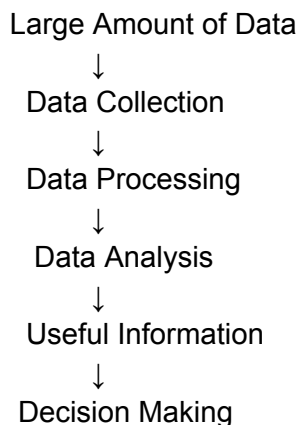
Analytics services are cloud-based services used to **collect, process, analyze, and interpret large amounts of data** to obtain useful information and support effective decision-making.

Cloud analytics services provide the required **storage, computing power, and analytical tools** through the cloud. Hence, organizations can analyze huge datasets without maintaining their own physical infrastructure.

Example

An e-commerce company can analyze millions of customer transactions to identify **best-selling products, customer preferences, sales trends, and total sales**.

General Working



Major Analytics Services

1. Amazon Elastic MapReduce (Amazon EMR)

Definition

Amazon Elastic MapReduce (EMR) is a cloud-based big data processing and analytics service provided by **Amazon Web Services (AWS)**. It is used to process and analyze large datasets using frameworks such as **Apache Hadoop** and **Apache Spark**.

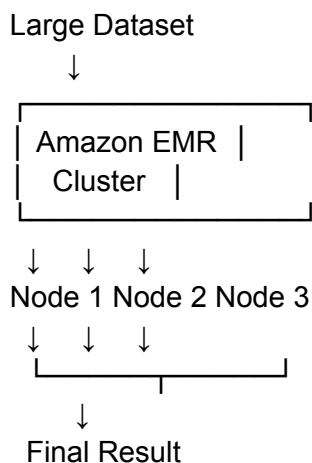
Features

- Processes large amounts of data.
- Uses distributed processing.
- Supports Hadoop and Spark.
- Provides scalable computing resources.
- Reduces the need for physical servers.
- Suitable for big-data applications.

Example

An e-commerce company can use Amazon EMR to process millions of customer transactions and identify **sales patterns and popular products**.

Diagram



Key point:

Amazon EMR = AWS + Big Data Processing

2. Google MapReduce

Definition

Google MapReduce is a distributed programming model developed by Google for processing **large datasets across multiple computers**.

It consists of two main operations:

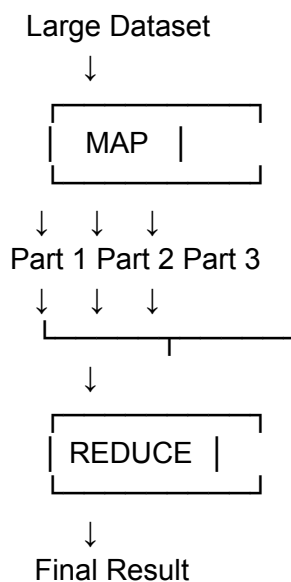
Map

The **Map operation** divides the input data into smaller parts and processes them independently.

Reduce

The **Reduce operation** combines the intermediate results produced by the Map operation to obtain the final result.

Working



Example

To count words in millions of documents:

- **Map:** Counts words from different parts of the documents.
- **Reduce:** Combines all the word counts to produce the final count.

Advantages

- Handles huge datasets.
- Supports parallel processing.
- Divides complex tasks into smaller tasks.
- Reduces processing time.

- Suitable for distributed computing.

Key point:

Google MapReduce = Map + Reduce + Distributed Processing

3. Google BigQuery

Definition

Google BigQuery is a **fully managed, serverless cloud data warehouse and analytics service** provided by Google Cloud. It is used to **store and analyze very large datasets using SQL queries**.

Features

- Serverless service.
- Supports SQL queries.
- Handles very large datasets.
- Provides fast data analysis.
- Highly scalable.
- No need to manage physical servers.

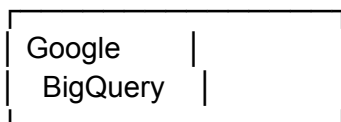
Example

A retail company can store millions of sales records in BigQuery and use SQL queries to identify:

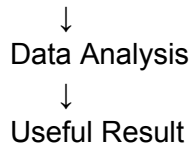
- Total sales
- Best-selling products
- Monthly sales
- Customer purchasing patterns

Diagram

Large Dataset



SQL Query



Key point:

Google BigQuery = Huge Data + SQL Analysis

4. Windows Azure HDInsight

Definition

Azure HDInsight is a cloud-based big data analytics service provided by **Microsoft Azure**. It is used to process and analyze large datasets using technologies such as **Apache Hadoop** and **Apache Spark**.

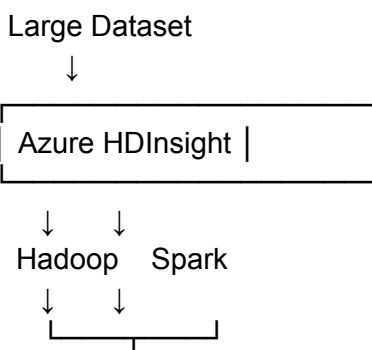
Features

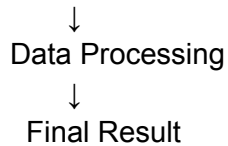
- Processes large amounts of data.
- Supports Hadoop and Spark.
- Provides scalable data processing.
- Runs big-data applications in the cloud.
- Reduces infrastructure management.
- Suitable for big-data analytics.

Example

A company can use Azure HDInsight to analyze millions of **website log records** and identify user activities, number of visitors, and website usage patterns.

Diagram





Key point:

Azure HDInsight = Azure + Hadoop/Spark + Big Data

Comparison

Service	Provider	Main Use	Key Technology
Amazon EMR	AWS	Big-data processing	Hadoop, Spark
Google MapReduce	Google	Distributed processing	Map + Reduce
Google BigQuery	Google Cloud	Large-scale data analysis	SQL
Azure HDInsight	Microsoft Azure	Big-data analytics	Hadoop, Spark

Conclusion

Analytics services play an important role in cloud computing by providing **scalable and cost-effective solutions for processing and analyzing large amounts of data**. **Amazon EMR, Google MapReduce, Google BigQuery, and Azure HDInsight** help organizations convert raw data into useful information and support **better decision-making**.